AI-ASSISTED NON-INVASIVE TURTLE NEST DETECTION

AND SAFE RELOCATION SYSTEM

THE CONSERVATION CHALLENGE

NEST DETECTION CHALLENGE

Turtle nests are buried **1–1.5 feet** beneath the sand. Current methods rely on **manual probing** with sticks, which is:

- Invasive and destructive
- Prone to damaging eggs
- Highly labor intensive

EGG RELOCATION CHALLENGE

Relocation must occur within **2–6 hours**. Minor changes in egg orientation can cause catastrophic losses.

UP TO 40%

MORTALITY RATE

Caused by manual handling and orientation stabilization failures.

STRATEGIC OBJECTIVES

01
NON-INVASIVE DETECTION
Eliminate physical probing to prevent accidental damage to buried egg clusters.

02
MULTI-SENSOR FUSION
PRECISION LOCALIZATION
Integrate Thermal, GPR, and LiDAR data for high-confidence nest identification.

03
ORIENTATION CONTROL
Active stabilization to maintain 0° tilt, preserving embryo viability during transport.

04
DAMAGE MINIMIZATION
Automated excavation and vibration-isolated transport to reduce human error.

05
IMPROVED SURVIVAL
Targeting a significant increase in hatchling success rates across global beaches.

GOAL
SCALABLE TECHNOLOGY
Creating a modular system deployable in diverse coastal environments.

THERMAL IMAGING DETECTION

WORKING PRINCIPLE

Thermal cameras detect **infrared radiation** emitted by objects, converting it into detailed thermal maps. In turtle conservation, this identifies:

- **Disturbed Sand**: Freshly moved sand has different cooling rates.
- **Moisture Retention**: Altered moisture levels create thermal anomalies.
- **Egg Chambers**: Subtle temperature variations from buried clusters.

Ideal for **drone-based deployment** to scan large beach areas rapidly.

ADVANTAGES

- Completely non-contact and non-invasive.
- Operates effectively during nighttime.
- Rapid scanning of expansive nesting sites.

LIMITATIONS

- Weak temperature contrast during hot daytime.
- Requires sensor fusion for precise localization.
- Surface-level indicators only.

GROUND PENETRATING RADAR

WORKING PRINCIPLE

GPR transmits **electromagnetic pulses** into the sand. Signals reflect back when they encounter materials with different dielectric properties:

- Air Cavities (Egg Chambers)
- Moisture Layers
- Egg Clusters
- Altered Sand Density

ADVANTAGE

Provides high-resolution subsurface mapping without any physical disturbance.

NEST LOCALIZATION

The buried egg chamber alters the internal structure of the beach, allowing the system to estimate:

DEPTH & GEOMETRY

By analyzing signal reflections, the AI identifies the **exact coordinates** of the egg cluster, enabling precision excavation by the relocation rover.

Applications: Utility mapping, Archaeology, and now, Non-Invasive Conservation.

SUPPLEMENTARY SENSING

MOISTURE MAPPING

SURFACE ANALYSIS

Scans surface moisture distribution patterns altered by underground nest construction.

CONFIDENCE BOOST

Acts as a low-cost supplementary layer to validate findings from Thermal and GPR sensors.

LIMITATION

Intended as a supporting sensor; cannot independently detect nests at 1–2 feet depth.

LIDAR ANALYSIS

3D TERRAIN MAPPING

Generates high-resolution topological maps using rapid laser pulse reflections.

PRECISION DETECTION

Identifies millimeter-scale surface depressions and disturbed sand regions indicative of nesting.

AUTONOMOUS NAVIGATION

Provides essential spatial data for the relocation rover's path planning and obstacle avoidance.

AI-BASED SENSOR FUSION

THERMAL IMAGING
Heat anomalies & cooling rates

GPR SIGNATURES
Subsurface dielectric changes

MOISTURE LEVELS
Surface moisture distribution

LIDAR TOPOLOGY
3D terrain & surface depressions

INTELLIGENT FUSION FRAMEWORK

Machine learning models analyze cross-sensor correlations to filter environmental noise and identify unique nesting signatures.

PROCESSING LAYER: ESP32 / CLOUD AI

RESULTING OUTPUT

98.4%

NEST LOCALIZATION CONFIDENCE

Enables precise deployment of the relocation rover.

RELOCATION ROVER SPECS

PHYSICAL DIMENSIONS

LENGTH	1.5 – 2.0 Meters
WIDTH	0.8 – 1.0 Meter
HEIGHT	1.0 – 1.2 Meters
WEIGHT	Below 150 KG

MOBILITY SYSTEM

Rubber tracks with ultra-low ground pressure for stable operation on loose sand.

CORE COMPONENTS

ROBOTIC ARM

Multi-axis manipulator for precision excavation.

TRANSPORT CHAMBER

Stabilized environment for egg safety.

SENSOR MODULE

Integrated GPR and LiDAR for local navigation.

CONTROL HUB

ESP32-based real-time processing unit.

The rover is designed to minimize beach disturbance while ensuring the safe extraction and transportation of endangered turtle nests.

EGG STABILIZATION SYSTEM

TRANSPORT CHAMBER

DIMENSIONS

2 FT × 2 FT

A specialized, climate-controlled environment designed to house turtle eggs during the critical relocation window.

MODULAR DESIGN

Features removable trays that allow for quick, safe transfer of egg clusters without direct handling.

SHOCK ABSORPTION

Integrated high-density foam and spring-loaded supports to neutralize mechanical impacts.

VIBRATION ISOLATION

Damping mechanisms to prevent engine or terrain vibrations from reaching the embryos.

ENVIRONMENTAL PROTECTION

Shields eggs from UV exposure and maintains stable internal temperature and humidity.

ACTIVE ORIENTATION PRESERVATION

STEP 01: SENSOR LAYER

IMU DETECTION

A 6-axis **MPU6050 Inertial Measurement Unit** continuously monitors the egg cradle.

- Real-time Pitch & Roll
- Angular Velocity
- Vibration Analysis

STEP 02: CONTROL LAYER

PID ALGORITHM

The **ESP32 Microcontroller** processes sensor data to calculate correction values.

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d de/dt$$

Minimizes orientation error with high-frequency feedback loops.

STEP 03: ACTUATION LAYER

SERVO STABILIZATION

High-torque **Servo Motors** drive a two-axis gimbal platform.

- Instant Tilt Correction
- Shock Compensation
- Mechanical Isolation

0° TILT

STRICT ORIENTATION TARGET

OPERATIONAL SYSTEM WORKFLOW

STEP 01

BEACH SURVEY

Initial scanning using Thermal Imaging, LiDAR, and Moisture sensors to map the beach.

STEP 02

IDENTIFY SITES

AI models flag potential nesting locations based on surface anomalies and thermal data.

STEP 03

GPR VERIFICATION

Ground Penetrating Radar confirms the presence and depth of underground egg chambers.

STEP 04

SENSOR FUSION

Integrated data analysis determines the exact 3D coordinates for the relocation team.

STEP 05

ROVER DEPLOYMENT

Relocation rover is dispatched to the precise GPS coordinates of the verified nest.

STEP 06

EXCAVATION

Robotic arm carefully extracts the nest and transfers it to the stabilization chamber.

STEP 07

PID STABILIZATION

Active control system maintains 0° tilt to preserve natural egg orientation during transit.

STEP 08

SAFE TRANSPORT

Vibration-isolated transport of the nest to the protected hatchery facility.

STEP 09

REBURIAL

Eggs are safely reburied under controlled conditions to maximize hatchling survival.

IMPACT & CONCLUSION

EXPECTED IMPACT

MORTALITY REDUCTION

Eliminates invasive probing and preserves egg orientation to maximize hatchling survival.

CONSERVATION MODERNIZATION

Shifts manual labor to high-tech, data-driven precision using AI and robotics.

SCALABLE SOLUTIONS

A modular blueprint deployable across global nesting beaches for biodiversity protection.

TECHNOLOGY SERVES AS A POWERFUL ALLY IN PRESERVING BIODIVERSITY AND SUPPORTING LONG-TERM ECOLOGICAL HEALTH.

NEXT STEPS

Prototype field testing and deployment on active nesting beaches to validate system performance.